Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with 2+ years building **cloud data pipelines**, **REST APIs**, and distributed applications using **Python**, **Java**, **AWS**, SQL, Docker, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Built **ETL data pipelines** with FastAPI, Supabase, SQL Server, and cloud storage, processing 5K+ monthly jobs through incremental loads, reconciliation, error handling, and **CI/CD**, achieving 99%+ production reliability.
- Architected multi-tenant **Azure SQL** and Blob Storage schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through indexed queries, normalization, and access-pattern optimization.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Designed a concurrent **data-processing pipeline** with Python multithreading and secure PDF, DOCX, and ZIP ingestion through AWS RDS and REST APIs, processing 1K+ files per batch reliably.
- Implemented an **AWS** production backend serving 150+ instructors and 10K+ daily grading requests, using CI/CD-driven workflows to reduce feedback turnaround 60% while preserving stable peak-load API performance.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *LA, California*

- Built a Python Decision Transformer pipeline processing 100K+ autonomous-driving trajectory logs, improving long-horizon planning quality 18% through reproducible data preparation, model training, simulation evaluation, and experiment analysis.
- Architected a spatial-temporal encoder combining HD-map geometry with historical agent trajectories through multi-head attention, improving closed-loop planning stability 15% while supporting systematic testing across complex driving scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Optimized large-scale **API integrations** using OOP patterns, **AWS Glue**, and S3, improving response times 30%; led scalable ETL workflows that raised pipeline efficiency 68% and reduced technical debt 20%.
- Engineered cross-platform data-sharing microservices with automated builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito, supporting reliable releases across three operating systems in Agile delivery.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Developed a MERN resource platform with WebSockets, **REST APIs**, and microservices, synchronizing 2M+ records across locations while reducing conflicts 33% through resilient real-time data flows and review practices.
- Automated **Docker** packaging and Jenkins CI/CD deployments, cutting release time 50%, while Azure SQL and Google Analytics insights reduced resource waste 42% and increased lead conversion 15%.

Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C++, C#
Data Engineering: ETL, Apache Airflow, Apache Spark, Hive, AWS Glue, Data Pipelines, Batch Processing, Data Validation
Backend & Databases: REST APIs, FastAPI, Flask, Node.js, PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, Redis
Cloud & Infrastructure: AWS, Azure, GCP, S3, RDS, ECS/Fargate, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD
Engineering: OOP, Microservices, Distributed Systems, Big Data, System Design, Unit Testing, Integration Testing, Git, Agile/Scrum

Projects

RokuAdFlow | Python, Airflow, Hive, Advertising ETL

- Built and tested an **Airflow-compatible advertising ETL** pipeline that validates and deduplicates events, aggregates campaign metrics, writes atomic date partitions, and generates **Hive-compatible** external-table DDL for analytics workloads.

Chat2SQL Code | Java, TypeScript, SQL, Docker

- Built an AI-powered SQL development workspace across a **Java** backend and TypeScript frontend, supporting 10+ databases with Dockerized workflows for schema exploration, query authoring, and cross-platform data management.

Big-Brain Live | GCP, PostgreSQL, REST APIs

- Engineered a cloud data-ingestion and indexing pipeline processing 50K+ records, implementing incremental updates, metadata persistence, and performance tuning to reduce retrieval latency 40% across scalable document analytics workloads.